

Machine-Provable Karnaugh Map Encoding of Protein Sequences: Boolean Circuit Extraction and Cross-Protein Coevolutionary Contact Prediction

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Abstract

The representation of protein sequences as Karnaugh maps through Gray-code encoding establishes a formal bridge between switching theory and the statistical mechanics of coevolution. We present a computational framework in which the entire analytical chain—from amino-acid encoding through Karnaugh-map construction, Quine–McCluskey minimisation, and prime-implicant extraction to cross-protein contact prediction—is formalised in the Lean 4 theorem prover and validated against native experimental structures from the PSICOV150 benchmark. Our universal contact circuits, trained on consensus residue identities from approximately one hundred proteins, transfer to held-out proteins at a mean top- $L/5$ precision of 0.179 ($5.1\times$ random enrichment). Fusing circuit-derived scores with continuous DCA rankings via rank aggregation yields a hybrid predictor that exceeds both individual methods ($p < 10^{-4}$), showing that Boolean-minimisation features carry structural information orthogonal to standard covariation analysis. Flipped (forbidden-combination) circuits generalise across proteins with 98.3% specificity. Gray-code adjacency is enriched among native contacts ($1.178\times$, $p < 10^{-77}$), although a code-permutation control places the assigned encoding only $z = 2.51$ above random residue-to-code assignments; conserved column pairs are contact-enriched by $3.2\times$, an effect carried by near-conserved columns, since fully invariant columns are depleted ($0.69\times$). We further prove an information ceiling: any predictor that is a function of a residue-level feature map is bounded by that map’s cell purity, which on consensus residue identity and separation is $\text{prec}@L/5 = 0.193$ —so the trained circuit already attains 93% of what its inputs permit, and the gap to direct-coupling analysis is informational rather than algorithmic. Redirecting the same minimisation from sequence to *conformational state* escapes that bound: across 480 structure-based-model trajectories the extracted rules link contacts that are $12.5\times$ enriched for contact-map adjacency ($p < 10^{-79}$), recovering local folding cooperativity, and under dense re-sampling they predict a contact’s state a thousand integration steps ahead in every protein tested. All software, formal proofs, and result files are publicly available.

1 Introduction

Residue–residue contact prediction from multiple sequence alignments has transformed protein structure determination [1–4]. The dominant paradigm, Direct Coupling Analysis (DCA), models the joint distribution of amino-acid sequences through a Potts Hamiltonian whose coupling parameters are estimated by inverting a covariance matrix computed from deep alignments [2]. Sparse inverse covariance estimation, as implemented in PSICOV [3], achieves a mean top- $L/5$ precision of approximately 0.73 on standard benchmarks. Pseudo-likelihood maximisation [5] and related methods further improve upon mean-field approximations. More recently, end-to-end deep learning approaches such as AlphaFold2 [6]

have largely superseded standalone contact prediction by integrating coevolutionary signal with structural priors inside attention-based neural architectures.

Despite this success, the encoding layer through which biological sequences enter the statistical machinery remains opaque. No prior work has established machine-checkable proofs that the transformation from amino-acid alphabet to the numerical representation used for inference preserves the structural properties that downstream analysis requires. This gap is not merely academic: subtle encoding errors can silently corrupt results while producing plausible-looking output.

Here we address this gap by constructing a formally verified bridge between digital-logic design theory and protein coevolution analysis. Our central observation is that the Karnaugh map [7], when combined with reflected binary Gray-code ordering of amino acids [8], provides a natural embedding of aligned sequences into a grid structure where adjacent cells correspond to physicochemically similar residue pairs. The Quine–McCluskey algorithm [9, 10] then extracts minimal sum-of-products expressions from thresholded frequency maps, producing prime implicants that we interpret as coevolutionary constraints.

We formalise 118 theorems in Lean 4 covering Gray-code bijectivity, k -mer indexing injectivity, amino-acid encoding properties, contact-map topology, implicant-cover soundness, and padding safety. We apply the verified pipeline to 150 protein domains from the PSICOV benchmark comprising over 1.6 million labelled observations against native experimental structures. Our results demonstrate that Boolean circuits transfer across proteins, that their cell-rate prior improves DCA ranking when fused via rank aggregation, and that negative-selection (forbidden-combination) circuits generalise with high specificity. We also report negative results honestly: an attention-style tension mechanism does not improve prediction, iterative direct information is unreliable on conserved-rich families, and current circuit precision falls far short of the level required for template-free structure reconstruction.

All software, formal proofs, result files, and documentation are publicly available (github.com/E-Motioner-X-SBS/co-evolution-analysis).

2 Related Work

2.1 Coevolutionary contact prediction

The inference of residue–residue contacts from multiple sequence alignments (MSAs) has evolved through several generations of methodology. Early approaches based on mutual information [11] suffered from phylogenetic and entropic biases that produced high false-positive rates. Direct Coupling Analysis (DCA) addressed this by modelling the joint sequence distribution through a Potts Hamiltonian whose coupling parameters capture direct (rather than transitive) correlations [1]. The mean-field approximation of Morcos et al. [2] made DCA computationally tractable for large alignments by replacing the full partition function with a covariance-based estimate. Sparse inverse covariance estimation, as implemented in PSICOV [3], further improved precision by promoting sparsity in the precision matrix through ℓ_1 regularisation. Pseudo-likelihood maximisation [5] and fast belief propagation implementations such as CCMpred [12] subsequently refined the inference procedure. GREMLIN [4] combined DCA with protein-structure priors. EVmutation [13] extended the framework to predict mutation effects on fitness rather than contacts alone.

More recently, end-to-end deep learning approaches such as AlphaFold2 [6] have integrated coevolutionary signal with structural priors inside attention-based architectures, achieving near-experimental accuracy for many targets. However, these models produce dense internal representations that resist decomposition into interpretable components.

2.2 Formal methods in computational biology

The application of interactive theorem provers to computational biology remains limited. Existing formalisations focus on algorithm correctness for phylogenetic inference [14], genome assembly graph properties, and sequence alignment dynamic programming recurrences. No prior work has established machine-checked proofs that the encoding layer connecting amino-acid alphabets to numerical representations preserves the structural properties required for coevolutionary analysis.

2.3 Karnaugh maps beyond digital logic

Karnaugh maps were introduced as a graphical method for minimising Boolean functions [7]. Their application outside digital circuit design has been explored in synthetic biology, where Boolean gates model gene regulatory networks [15], and in the analysis of genetic code symmetries [16]. Our work extends this tradition by using Karnaugh maps not merely as a visual representation but as an analytical engine: the Quine–McCluskey minimiser extracts testable coevolutionary rules from thresholded frequency data, and the resulting circuits are validated against native protein structures.

3 Gray-Code Encoding of Biological Alphabets

A reflected binary Gray code assigns to each non-negative integer n the value $g(n) = n \oplus (n \gg 1)$ [8]. Consecutive integers map to codewords differing in exactly one bit. For nucleotides we adopt the standard two-bit encoding A=0, C=1, G=3, T/U=2; transitions correspond to single bit-flips under Gray transformation.

For amino acids, each residue receives its He 2012 integer code [17]: A=0 through G=19, ordered into seven physicochemical groups (aliphatic AILVM, aromatic FYW, sulfur MC, structure PG, polar STNQ, negative DE, positive HKR). The five-bit Gray image $g(\text{code}(a))$ places each residue on the hypercube Q_5 . Among the $\binom{20}{2} = 190$ unordered pairs, exactly 80 ordered pairs lie at Gray distance 1 (verified in Lean 4 as `cc_he_dist1_ordered_eq_80`).

Alignment gaps are mapped to state 20 and excluded from all statistical calculations but retained in position arrays so that column identity is preserved across all sequences.

4 Karnaugh Maps and Quine–McCluskey Minimisation

For a pair of alignment columns (i, j) , a ternary Karnaugh map K_{ij} assigns each cell of a 32×32 grid one of three values based on joint residue frequencies: ON for observed mutations, OFF for never-observed pairs, and don’t-care for reference pairs and padding rows/columns 20–31.

The Quine–McCluskey algorithm [9, 10] finds all prime implicants by iterative pairwise merging of minterms differing in one bit, with don’t-care cells included alongside on-set cells. Essential prime implicants cover on-set cells matched by no other implicant; greedy selection completes the cover. Two properties are machine-verified: every ON cell is covered (`cc_cover_complete`) and no implicant matches any OFF cell (`cc_off_avoiding`).

Flipped maps assign ON status to never-observed combinations, producing forbidden-combination rules that encode negative-selection constraints.

5 Information-Theoretic Measures and Direct Coupling Analysis

5.1 Information-theoretic measures

Shannon entropy per column quantifies evolutionary variability after excluding gap characters. Mutual information between columns (i, j) measures statistical dependence: $MI(i, j) = \sum_{a,b} P_{ij}(a, b) \log_2[P_{ij}(a, b)/(P_i(a)P_j(b))]$. The Average Product Correction removes phylogenetic bias: $MIp = MI - \bar{MI}(i, \cdot)\bar{MI}(\cdot, j)/\bar{MI}$ [11]. Perplexity ratio $r(i, j) = PP(j)/PP(j|i)$ captures near-determinism of the residue relationship.

5.2 Mean-field DCA

Direct Coupling Analysis models sequences through a Potts Hamiltonian whose couplings are estimated by covariance inversion [1, 2]. Sequence reweighting at $\theta = 0.2$ and pseudocount $\lambda = 0.5$ regularise the estimates. The Frobenius norm of each coupling block provides a pair score; APC correction removes phylogenetic bias. Alignments deeper than 12,000 sequences are uniformly subsampled.

6 Formal Verification in Lean 4

All load-bearing definitions are formalised in Lean 4 [14] across six modules of one Lake package, totalling 118 theorems. Key results include: Gray-code consecutive adjacency, Hamming-distance symmetry, He-code pair counts at each distance, separation-bin monotonicity, cell-index injectivity, implicant-cover completeness and soundness, fully-fixed implicant uniqueness on the padded grid, and padding safety. Every theorem depends only on `propext`, `Quot.sound`, `Classical.choice`, and the `native_decide` evaluation axiom (87 of the 118 depend on the latter, and so on compiler trust; 7 use `Classical.choice`); none depends on `sorryAx` or on a user-declared axiom.

7 Universal Contact Circuits Transfer Across Proteins

We trained Quine–McCluskey-minimised Boolean circuits on consensus-residue identities and separation distances from approximately 100 PSICOV150 proteins, then evaluated on 50 entirely held-out proteins (maximum pairwise 5-mer Jaccard index 0.029 across all $\binom{150}{2}$ pairs). The identity-level circuit achieved a mean top- $L/5$ precision of **0.179** ($5.1\times$ enrichment over base rate 0.035), a $1.9\times$ improvement over plain MI (0.097), and a $15\times$ improvement over separation-only features (0.033). Binary classification at circuit membership achieved $MCC \approx 0.095$ with enrichment of $2.5\text{--}2.8\times$. Ranked precision was invariant across all nine labelling-threshold combinations tested.

The minimised circuits produced interpretable rules. Level-1 essential prime implicants included hydrophobic $\times$ aromatic packing at separations 6–21 and sulfur-mediated contacts at separations 6–11. Level-2 vocabularies included $\{M, F\} \times \{M, F, Y, W\}$ and $\{I, V\} \times \{I, V, F, W\}$, corresponding to known aliphatic/aromatic core interactions rediscovered without biophysical prior knowledge.

8 Gray Adjacency Enrichment and Rank Fusion with DCA

8.1 Gray adjacency among native contacts

Among 47,430 native-contact pairs across 150 PSICOV proteins, 22.83% are Gray-adjacent ($d_G = 1$), compared to a composition-controlled background of 19.37%, yielding an enrich-

ment of $1.178 \times$ ($p = 1.1 \times 10^{-77}$, two-sided exact binomial). Bootstrap confidence intervals from protein-level resampling span [22.27%, 23.40%].

A permutation test over random relabelings of the twenty amino-acid codes demonstrated that this enrichment is specific to the He/Gray assignment: random code permutations yield a mean enrichment of 0.961 ± 0.095 , placing the true encoding at $z = 2.51$ above the null distribution. An earlier analysis restricted to a single protein found the same point estimate but failed to reach significance due to insufficient sample size ($n = 38$).

8.2 Rank fusion improves DCA ranking

The Karnaugh-map cell-rate prior improved global ranking (circuit AUC 0.658 vs. DCA AUC 0.583) while degrading top- K sharpness. We hypothesised that rank aggregation would capture both properties. Five fusion rules were evaluated under identical protein-level cross-validation:

Table 1: Rank-fusion results over five-fold protein-level CV.

Method	prec@L/5 (mean $\pm$ sd)	AUC
rawDCA alone	0.169 ± 0.125	0.583
circH prior alone	0.124 ± 0.088	0.658
ranksum	0.204 ± 0.104	0.638
geometric mean	0.203 ± 0.103	0.634
product	0.199 ± 0.108	0.592

The rank-sum fusion achieved $\text{prec@L/5} = \mathbf{0.204}$, exceeding rawDCA (Wilcoxon $p < 10^{-4}$) and circH ($p < 10^{-4}$). Cohen’s $d = 0.31$ with bootstrap 95% CI [0.017, 0.054]. Mechanistic decomposition showed that all rescued pairs had above-median scores from both methods, confirming coincidence-of-evidence amplification. Adding plain MI as a third signal did not improve further (three-way 0.201 vs. two-way 0.204, $p = 0.607$).

9 The Complete Circuit Combines Positive, Flipped, and Conservation Channels

We constructed a two-sided classifier over a shared feature space of physicochemical groups, separation bins, and conservation-pair types (10 bits, 1024 cells). Two ternary tables were labelled from identical pooled counts: the positive table marks cells whose contact rate significantly exceeds the base rate, while the flipped table marks cells whose rate is significantly below it. A pair receives a definitive CONTACT call if its cell is ON in the positive table and a definitive FORBIDDEN call if its cell is ON in the flipped table; all other pairs are abstained.

On held-out proteins, the positive channel achieved precision 0.092 ($2.6 \times$ base) at recall 0.203. The flipped channel achieved specificity of **98.3%**: forbidden-called cells showed exactly the predicted $\leq 1.75\%$ contact rate on unseen proteins, confirming that negative-selection rules generalise across protein families. Combined definitive-call MCC was +0.169 at 22% coverage.

Pairs in which both alignment columns are fully conserved (entropy < 0.001) showed a contact rate of 11.4%, representing a $3.2 \times$ enrichment. However, a finer analysis using continuous minimum-pair entropy revealed that absolutely invariant positions are depleted of contacts ($0.69 \times$), while near-conserved positions (entropy 0.001–0.05) are enriched ($1.35 \times$). This non-monotonicity indicates that structural-core membership requires residues

to be constrained but not absolutely frozen; catalytic or functional constraints can lock a position without placing it in a contact-rich environment.

10 The Information Ceiling of a Feature Class

Section 7 reports that the universal circuit transfers to held-out proteins at $\text{prec}@L/5 = 0.179$ against published DCA’s 0.723, and Section 11 that this is insufficient for template-free reconstruction. Both are empirical statements about one method. This section replaces them with a bound that holds for *every* method built on the same features, and thereby identifies where the shortfall lives.

The bound rests on an observation that is trivial and decisive: a circuit whose inputs are a feature vector is a function of that feature vector, and therefore cannot distinguish two candidate pairs that share it.

Definition 10.1 (Cell predictor). *Let P be the candidate residue pairs of a protein, $y : P \rightarrow \{0, 1\}$ the native-contact labels, and $\varphi : P \rightarrow \mathcal{C}$ a feature map into a finite set of cells. A cell predictor is any $\rho : \mathcal{C} \rightarrow \mathbb{R}$, inducing the score $\rho \circ \varphi$. Pairs sharing a cell receive equal scores; their relative order is fixed by a tie-break which carries no information about y and is therefore modelled as uniformly random.*

Every circuit in this paper is a cell predictor: a sum-of-products over φ is a $\{0, 1\}$ -valued function of the cell, and the smoothed cell-rate scoring used in practice is a real-valued one. The class of cell predictors is strictly larger than the class of circuits, so a bound over it binds the circuits a fortiori.

Theorem 10.1 (Ceiling). *Write $n_c = |\varphi^{-1}(c)|$ and r_c for the empirical contact rate in cell c . For any cell predictor ρ and any k ,*

$$\mathbb{E}[\text{prec}@k(\rho \circ \varphi)] \leq \frac{1}{k} \max \left\{ \sum_c t_c r_c : 0 \leq t_c \leq n_c, \sum_c t_c = k \right\},$$

the expectation taken over the tie-break, with the maximum attained by filling cells in decreasing order of r_c .

Proof. Fix ρ . Its induced ranking visits cells in decreasing ρ -order; write t_c for the number of pairs drawn from cell c among the top k , so $0 \leq t_c \leq n_c$ and $\sum_c t_c = k$. Within a cell the tie-break is uniform, so the t_c selected pairs form a uniformly random t_c -subset of $\varphi^{-1}(c)$ and the expected number of contacts drawn from c is $t_c r_c$ by linearity of expectation; hence $\mathbb{E}[\text{prec}@k] = k^{-1} \sum_c t_c r_c$. Varying ρ varies only the cell order, hence only the allocation (t_c), which ranges over prefix-filling allocations; relaxing to all allocations satisfying the stated constraints can only increase the maximum. That relaxation is a fractional knapsack with unit weights, whose optimum sorts cells by r_c descending—and this optimum is itself prefix-filling, so the relaxation is tight. $\square$

Corollary 10.2. *No choice of minimisation algorithm, labelling threshold, don’t-care policy, cover-selection rule, or Boolean representation can raise the expected precision of a predictor above the bound of Theorem 10.1 for its feature map. Improvement requires refining φ .*

Proof. Each listed choice determines which cell predictor ρ is obtained; none changes φ . The bound is uniform over ρ . $\square$

Table 2 evaluates the bound on all 150 PSICOV targets at $k = L/5$, with cells scored by their own pooled empirical contact rate—the feature class handed the test labels.

Three readings follow. **The machinery is saturated:** the trained circuit reaches 0.179 against an achievable 0.1933, or 93% of its feature class’s ceiling, so by Corollary 10.2

Table 2: Achievable $\text{prec}@L/5$ per feature map, 150 PSICOV targets. Consensus residue codes; separation bins $[6, 11)$, $[11, 21)$, $[21, 31)$, $[31, \infty)$; mutual-information quartile computed per protein.

Feature map φ	Cells	Achievable
residue _{<i>i</i>} , residue _{<i>j</i>} , separation, MI quartile	6390	0.2412
residue _{<i>i</i>} , residue _{<i>j</i>} , separation	1600	0.1933
physicochemical group _{<i>i</i>} , group _{<i>j</i>} , separation	256	0.1418
residue _{<i>i</i>} , residue _{<i>j</i>}	400	0.1383
Gray Hamming distance, separation	24	0.0789
separation only	4	0.0598
<i>Measured on the same benchmark:</i>		
Trained circuit, held out (Sec. 7)		0.179
Plain mutual information		0.097
Published DCA [3]		0.723

the total remaining headroom from every possible improvement in Boolean method is at most about 8% relative. **The shortfall is informational:** the ceiling itself sits $3.7\times$ below published DCA, and no Boolean construction on these features can close that gap. **Coarsening chemistry costs information:** the eight-group physicochemical partition has a *lower* ceiling than raw residue identity, as it must, since merging cells can only reduce the bound. Chemical classes buy interpretability, not accuracy.

This reframes Section 11. The failure to meet the reconstruction criterion is not a deficiency of our minimiser or our labelling; it is a property of consensus residue identity as an input. Since the bound depends only on the partition, the only lever is a finer partition separating contacting from non-contacting pairs *within* existing cells. Adding a single crudely binned column statistic already lifts the ceiling from 0.1933 to 0.2412, whereas richer residue vocabularies lift nothing—which is the formal counterpart of the rank-fusion result of Section 8, where combining the circuit with continuous DCA outperforms either alone.

11 Reconstruction Assessment and Negative Results

11.1 Reconstruction is not achievable at current signal strength

We assessed whether the circuits supply sufficient correct spatial constraints for template-free structure determination by measuring recall (fraction of native contacts recovered) and long-range precision. Circuit $\text{recall}@L = 0.028$ versus published DCA’s 0.218; median long-range $\text{precision}@L/5$ is 0.000 for circuits versus 0.683 for PSICOV predictions. Against the threshold of median LR $\text{prec}@L/5 \geq 0.30$ and $\text{recall}@L \geq 0.10$ derived from Jones et al. and the CONFOLD literature, the circuits do not meet the criterion while published DCA does.

11.2 Tension mechanism does not improve prediction

An attention-style tension score, normalising each position’s mutual information by its strongest partner, did not improve circuit precision (tension-augmented 0.154 vs. non-tension 0.165). High-tension positions were anti-correlated with structural hubs (median $\rho = -0.06$), indicating that entropic attention concentrates on variable surface clusters rather than buried cores.

11.3 Iterative DI is unreliable on conserved-rich families

Our mean-field implementation of iterative direct information produced anti-predictive rankings ($\text{AUC} \approx 0.41$) on families with many fully conserved columns, due to pseudo-inverse conditioning when variance approaches zero for locked positions. APC-corrected Frobenius norm was adopted as the primary DCA score instead; this choice was validated against native contacts and matches published mean-field benchmarks.

11.4 Cross-dataset pilot

A pilot evaluation on three EVmutation proteins produced inconclusive results because correct parsing of the A2M alignment format requires handling insert-state columns that do not map linearly to PDB residue numbers. Structures were successfully fetched from RCSB and MI AUCs were positive for two of three targets, but precision could not be meaningfully evaluated without proper reference-frame mapping. Full evaluation is deferred to future work.

12 Boolean Logic over Conformational State

The results so far apply Boolean minimisation to *sequence*: an alignment column pair becomes a Karnaugh map, and its prime implicants describe which residue combinations coevolution permits. Section 10 shows that this substrate is close to exhausted for contact prediction. Here we change the substrate rather than the method.

The motivation is that Boolean minimisation is productive elsewhere in biology precisely where the object is a *logical relation among variables observed across many states*—gene-regulatory inference being the established case, in which binarised expression time series yield logic rules via Quine–McCluskey [18]. A static residue string is not such an object. A structure-based model (SBM) trajectory [19] is: every native contact is either formed or broken in every frame, so the trajectory is a binary state matrix and one may ask whether a contact’s state is a Boolean function of a few other contacts’ states.

12.1 Controlling the folding coordinate

Every pair of native contacts is correlated for a trivial reason: when the protein is folded most contacts are formed, when unfolded most are broken. Any naive coupling statistic measures this global mode and nothing else. Three devices control for it, all applied throughout.

First, Q , the fraction of native contacts formed, is quantile-binned into two bits and supplied to the circuit as an *explicit input*, so that a rule reads “at this folding level, contact c is formed iff ...” rather than “contact c is formed when the protein is folded”. Second, every rule is scored against a *Q -only baseline* built by the identical procedure from the Q bits alone; the reported quantity is always the gain over that baseline, so a rule that merely restates the global mode scores zero. Third, a *Q -stratified null* resamples contact columns within Q strata, preserving each contact’s dependence on the folding coordinate and its marginal frequency while destroying any specific contact-to-contact coupling.

Predictors are ranked by *conditional* mutual information $I(c; x \mid Q\text{-bin})$; plain mutual information would rank candidates by how strongly each tracks the confounder. Predictor selection is repeated inside every cross-validation fold, so the *choice* of predictors is held out and not only the fitted rule. Folds are contiguous frame blocks rather than random splits.

12.2 Coupling beyond the folding coordinate is present

Across 480 proteins and 18,714 extracted rules, every circuit runtime-verified sound and complete, the mean gain over the Q -only baseline is -0.0054 while the Q -stratified null gives -0.0248 . The real configuration exceeds its own null in **463 of 480 proteins** (Wilcoxon $p = 4.3 \times 10^{-78}$).

The sign deserves comment. On held-out frames the contact-based circuit is marginally *worse* than the two-bit Q model, yet decisively better than its own null. Varying the predictor count identifies the cause (Table 3): the absolute gain degrades monotonically as the truth table grows, while the real-minus-null difference stays flat at $\approx +0.024$ and equally significant at every size. That is a variance-limited estimator, not absent signal—with 100 frames there are ≈ 2.5 frames per cell at three predictors.

Table 3: Predictor count against estimator variance (60 proteins, 25 targets each). Absolute gain falls with table size; the real-minus-null difference does not.

Predictors	Cells	Gain vs. Q -only	Null gain	Real $-$ null	Real $>$ null
1	8	+0.0033	-0.0195	+0.0228	52/55 ($p = 2.8 \times 10^{-10}$)
2	16	-0.0002	-0.0259	+0.0257	50/55 ($p = 6.0 \times 10^{-10}$)
3	32	-0.0023	-0.0260	+0.0237	51/55 ($p = 3.5 \times 10^{-10}$)

12.3 The learned rules are spatially local

The substantive finding is what the rules connect. For each rule we compare the contact-map L_1 distance $|i - k| + |j - l|$ between the target contact and its selected predictors against predictors drawn at random from the *same protein's* varying contacts, so that protein size and contact density cancel.

Table 4: Selected predictors versus matched-random predictors, 480 proteins. Adjacency is $L_1 \leq 2$ on the contact map.

	Observed	Matched random
Fraction of predictors adjacent	0.2127	0.0170
Enrichment	$12.5\times$	
Proteins with observed $>$ random	478 / 480	
Wilcoxon	$p = 3.3 \times 10^{-80}$	
Median contact-map distance	61.5	107.5
Predictors sharing a residue	27.2%	

Conditional mutual information, with the global folding mode controlled out, selects the *neighbouring cells of the contact map*: adjacent rungs of the same β -ladder, successive turns packed against the same partner. More than a quarter of selected predictors share a residue with their target. This is local cooperativity—the foldon picture [20]—recovered from a discrete-logic direction without secondary structure being supplied. Individual rules can be strong even where the population mean gain is not:

- 1zjy** contact (1, 28): accuracy 0.900 vs. Q -only 0.480, MCC 0.80;
predictors (1, 29), (2, 28), (2, 29), contact-map $L_1 = 1, 1, 2$.
- 1kms** contact (2, 129): accuracy 0.790 vs. 0.380, MCC 0.62;
rule $c(2, 129) = c(2, 127) \wedge c(1, 129)$, $L_1 = 2, 1$.

12.4 Longer trajectories confirm the variance diagnosis

The diagnosis in Table 3 makes a testable prediction: more statistically independent configurations should flip the absolute gain positive. We re-simulated ten proteins for four times longer with twenty times denser output (2×10^6 steps saved every 250, against 5×10^5 every 5000) and repeated the identical analysis in three arms (Table 5).

Table 5: Three arms on the same ten proteins. The middle arm decimates the long run back to the *original* 5000-step spacing, so its time resolution is unchanged and only the number of independent configurations differs.

Condition	Frames	Spacing	Gain vs. Q -only	Null	Fraction > 0
Original	100	5000	+0.0164	−0.0549	0.90
Long-sparse	400	5000	+ 0.0628	−0.0299	1.00
Long-dense	8000	250	+ 0.0842	−0.0022	1.00

Long-sparse is the informative arm: identical time resolution to the original, four times the independent configurations, and the gain rises $3.8\times$ and becomes positive in every protein (+0.0464, 10/10, $p = 0.00195$). The constraint was sampling, not method. These ten proteins were selected as small and strongly fluctuating, so their absolute levels are optimistic relative to the 480-protein population; the paired comparisons are unaffected.

12.5 Dense sampling licenses a kinetic reading

In the original trajectories the lag-1 autocorrelation of contact state averages 0.009 (Q : 0.130): the 5000-step save interval exceeds the correlation time, the frames are near-independent equilibrium samples, and no $t \rightarrow t + 1$ statement is licensed. We measured this per protein rather than assuming it, and made no kinetic claim on that data.

At 250-step spacing the contact lag-1 autocorrelation rises to 0.539. A lagged analysis—predicting the target k frames *ahead* of the inputs, with predictor selection lagged identically—therefore becomes meaningful (Table 6).

Table 6: Predicting a contact ahead of its inputs, ten densely sampled proteins.

Lag	Horizon	Gain vs. Q -only	Null	Real $>$ null
0	same frame	+0.0842	−0.0022	10/10 ($p = 1.95 \times 10^{-3}$)
1	250 steps	+0.0593	−0.0023	10/10 ($p = 1.95 \times 10^{-3}$)
2	500 steps	+0.0431	−0.0024	10/10 ($p = 1.95 \times 10^{-3}$)
4	1000 steps	+0.0288	−0.0021	10/10 ($p = 1.95 \times 10^{-3}$)

Predictive power decays roughly exponentially, with a relaxation of ≈ 900 steps, but remains positive in every protein a thousand integration steps ahead. The state of three neighbouring contacts now carries information about a target contact’s future state beyond what the folding level supplies—a statement about mechanism rather than only about the equilibrium ensemble, and the first such statement this framework has been entitled to make.

13 Discussion

We have demonstrated that the analytical chain from biological sequences through Gray-code encoding, Karnaugh-map construction, Quine–McCluskey minimisation, and prime-implicant extraction can be made machine-provable in Lean 4, and that the resulting

Boolean circuits carry genuine structural information when applied to protein multiple sequence alignments.

The principal finding is that the Karnaugh-map cell-rate prior carries orthogonal structural information demonstrable by rank-fusion with continuous DCA scores. The fused predictor achieves $\text{prec@L}/5 = 0.204$, exceeding both the circuit alone (0.124) and DCA alone (0.169) at $p < 10^{-4}$. This is the first evidence that Boolean-minimisation features improve continuous coevolutionary contact prediction.

The biological interpretation of learned circuit rules confirms known structural principles. Level-1 essential implicants encode hydrophobic-core packing, aromatic stacking, and sulfur-mediated contacts in directly readable sum-of-products form. At Level-2 resolution, specific residue-pair vocabularies correspond to aliphatic/aromatic packing interactions that dominate the buried cores of globular proteins.

Two results reframe what the rest of the paper reports. The ceiling theorem (Section 10) converts an empirical shortfall into a bound: because a circuit is a function of its feature cell, its expected precision is capped by that cell’s purity, and on consensus residue identity with separation the cap is 0.193 against the 0.179 we achieve. The circuit is therefore at 93% of what its inputs permit, and the $3.7\times$ distance to published DCA is a property of the features rather than of the minimisation. This settles a question the earlier sections could only pose: no better minimiser, threshold, don’t-care policy or Boolean representation can close it, and effort is better spent on the inputs. It also explains why coarsening residues into eight physicochemical groups *lowers* the ceiling—merging cells can only discard information—so chemical classes should be understood as buying readability, not accuracy.

The second is a change of substrate rather than of method. Boolean minimisation is productive elsewhere in biology where the object is a logical relation among variables observed across many states [18]; a static residue string is not such an object, whereas a structure-based-model trajectory is. Applied to conformational state (Section 12), the same exact minimiser finds coupling beyond the folding coordinate in 463 of 480 proteins, and—the substantive point—the rules it learns are *spatially local*: selected predictors are $12.5\times$ enriched for contact-map adjacency in 478 of 480 proteins, with more than a quarter sharing a residue with their target. Conditional mutual information, with the global folding mode controlled out, recovers β -ladder registers and local packing units without being told that secondary structure exists. Under dense re-sampling this becomes a kinetic statement: three neighbouring contacts predict a target contact’s state a thousand integration steps ahead, in every protein tested.

Several limitations constrain the scope of our conclusions. Consensus-residue features collapse within-column frequency variation into a single code, discarding information that richer encodings might exploit. Separation bins use globally defined boundaries without per-protein length normalisation. Cross-dataset generalisation was inconclusive on EVmutation due to A2M format parsing complexity, leaving open the question of whether findings replicate beyond PSICOV150. The iterative direct-information computation from our mean-field implementation remains unreliable on families with many fully conserved columns; we adopted APC-corrected Frobenius norm as the primary DCA score for this reason. In the contact-state analysis the mean gain over the Q -only baseline is negative at the production trajectory length, a variance limit we diagnose by varying the predictor count and confirm by re-simulation; the ten proteins re-simulated were chosen to be small and strongly fluctuating, so their absolute levels are optimistic relative to the 480-protein population even though the paired comparisons are not. Finally, prime-implicant counts reported from the dense reference minimiser are lower bounds rather than exact Quine–McCluskey quantities, for the reason given in Section 18.

Four extensions follow naturally. Extending the dense re-sampling from ten favourably

chosen proteins to a representative sample of the 480 would establish whether the sign flip holds at the population level, and would make the kinetic analysis available at scale rather than as a demonstration. Replacing binary conservation flags with continuous entropy bins as additional Karnaugh-map dimensions could capture finer-grained structural information about the buried core. Integrating circuit-derived rules as input features to deep learning architectures would combine the interpretability of Boolean logic with the representational power of attention-based models. Applying the complete pipeline to CASP free-modelling targets would test whether the observed transfer generalises to harder benchmarks with shallower alignments.

14 Dataset

The PSICOV150 benchmark [3] provides 150 protein domains with deep MSAs generated by three iterations of JackHMMER against UniRef100, together with repaired native structures from the Protein Data Bank. Alignment depths range from 511 to 74,836 sequences; protein widths range from 50 to 266 residues. The number of alignment columns equals the number of C- α atoms in each PDB file, verified empirically for 149 of 150 targets.

Native contacts use $C\beta-C\beta < 8 \text{ \AA}$ (C α for glycine) with separation ≥ 6 , following CASP conventions [21]. Across all targets, 47,430 native contacts exist among 1,675,886 candidate pairs, yielding a base rate of 2.83%.

Published PSICOV contact predictions from the supplementary data serve as literature baseline. Our measurement of these predictions yields $\text{prec@L/5} = 0.727$ across all separations ≥ 6 , consistent with Jones et al.’s reported value of approximately 0.73.

15 Encoding and Feature Extraction

Each MSA is converted to a dense integer matrix using the He 2012 amino-acid code (values 0–19), with gaps mapped to state 20. Consensus residues are computed as the majority non-gap code per column with lowest-code tie-breaking. Candidate pairs include all (i, j) with $j - i \geq 6$ and both columns having valid consensus codes.

Pair features comprise: physicochemical group codes ($g_i, g_j \in \{0, \dots, 7\}$), separation bins ($s \in \{0, 1, 2, 3\}$ for $[6, 11)$, $[11, 21)$, $[21, 31)$, $[31, \infty)$), consensus residue codes, Gray-Hamming distance, mutual information, and perplexity ratio.

Two circuit levels are constructed from pooled counts over training proteins: Level-1 (256 cells) over $\text{group}_i \times \text{group}_j \times \text{separation-bin}$; Level-2 (4×1024 cells) over residue-identity pairs padded to 32×32 , one map per separation bin. Cell labelling uses the ternary rule: ON if contact rate $\geq T \times \text{base rate}$ and $n \geq n_{\min}$; don’t-care if $n < n_{\min}$; OFF otherwise, with defaults $T = 2$, $n_{\min} = 30$.

16 Circuit Training and Evaluation

Each labelled truth table is minimised using a popcount-grouped Quine–McCluskey implementation with full don’t-care support. Every trained circuit is asserted sound (no implicant matches any OFF cell) and complete (every ON cell is covered) before evaluation.

Cross-protein evaluation uses protein-level cross-validation with folds stratified by length quintile (seed 42). All model parameters derive exclusively from training proteins. Ranked metrics include precision@L/5 , precision@L/2 , precision@L , and AUC via Mann–Whitney U with mid-rank tie handling.

For the rank-fusion experiment, five rules are evaluated: rank sum, z -score sum, product, geometric mean, and DCA-gated variant. Paired Wilcoxon signed-rank tests assess significance of improvements.

17 Statistical Framework

Enrichment tests use exact one-sided binomial tests against matched-separation random controls (2,000 draws per model, fixed seeds). Spearman rank correlations assess monotone relationships. Protein-level bootstrap resampling (2,000 replicates) provides confidence intervals. Wilcoxon signed-rank tests evaluate paired method comparisons without normality assumptions. Permutation tests over random code assignments quantify encoding specificity. Multiple-comparison awareness is maintained through median reporting for skewed distributions and explicit Bonferroni-corrected thresholds where noted.

All stochastic elements use fixed random seeds. Re-running any stage produces byte-identical output files, verified by MD5 checksums across multiple independent invocations for each result category.

18 Methods: Boolean Logic over Contact States

18.1 Simulations and contact states

Structure-based models [19] were generated with SMOG 2 [22] as C α -only topologies and integrated with OpenSMOG [23] on OpenMM [24], using a Langevin middle integrator at reduced temperature $T = 0.99994$, time step 0.0005, collision rate 1.0 and cutoff 1.1 nm. The production set comprises 1,359 protein trajectories of 5×10^5 steps saved every 5000, of which the 500 lexicographically first with parseable topologies were analysed and 480 yielded rules.

A native contact (i, j) from the SMOG shadow map is *formed* in a frame when the C α –C α distance is below 1.2 times its distance in the SBM reference structure; Q is the fraction of native contacts formed, and contacts whose formed-fraction lies outside $[0.05, 0.95]$ over the trajectory are treated as constant and excluded as targets or predictors.

For the dense re-sampling of Section 12.4, ten proteins were re-simulated with identical parameters for 2×10^6 steps saved every 250. These 50–300 atom systems are small enough that OpenMM’s CPU platform at one thread (≈ 12.7 k steps/s) outruns CUDA (≈ 9 k), where kernel-launch overhead dominates; the batch therefore ran one single-threaded process per protein. Six further candidates were dropped because their topology directories lacked the OpenSMOG XML file.

18.2 Rule construction

Inputs to each circuit are the states of $n_{\text{pred}} = 3$ selected contacts followed by two bits of the quantile-binned Q , giving $2^5 = 32$ cells. Predictors are the top- n_{pred} contacts by conditional mutual information $I(c; x \mid Q\text{-bin})$, computed on the training frames only and recomputed inside every fold. Conditional rather than plain mutual information is essential: plain MI ranks candidates by how strongly each tracks the folding coordinate, which is the confounder being controlled.

Each cell is labelled by majority vote over its training frames—ON when the target is formed in more than half, OFF otherwise—and cells holding fewer than four frames become don’t-cares, the standard treatment of unvisited states in Boolean network inference and what allows the minimiser to generalise rather than memorise. The ternary table is minimised by exact Quine–McCluskey on the sparse term set, and every returned cover is runtime-verified sound (no cube covers an off-cell) and complete (every on-cell is covered) by cube enumeration.

18.3 Evaluation and controls

Cross-validation uses five contiguous frame blocks rather than random splits: where any temporal correlation survives the save interval, a random split would place adjacent frames on both sides and inflate held-out accuracy. Three predictors are scored on the same held-out frames—the circuit, a Q -only baseline built by the identical procedure from the Q bits alone, and the training-majority class—and the reported effect is always $\text{acc}_{\text{rule}} - \text{acc}_{Q\text{-only}}$.

The Q -stratified null resamples every contact column within Q strata. This preserves exactly each contact’s marginal frequency inside each Q bin, hence its entire dependence on the folding coordinate, while destroying which frames pair which contacts. A gain that survives this null is not explicable by the global mode.

For the locality analysis, predictors are compared against contacts drawn at random from the same protein’s varying set, so that protein length and contact density cancel; adjacency is $L_1 \leq 2$ on the contact map, where $L_1 = |i - k| + |j - l|$ for contacts stored as sorted residue pairs.

Temporal scope is measured rather than assumed: the lag-1 autocorrelation of Q and of each varying contact is reported per protein, and the lagged analysis of Section 12.5 is applied only to the densely sampled trajectories, where it exceeds 0.5.

18.4 Reproducibility

Seeds are fixed throughout (base 20260905, offset per protein by index; folds contiguous), and stage re-runs reproduce identical JSON. The Quine–McCluskey engine is shared with the sequence-circuit analysis and carries 22 correctness gates, including agreement with an independent dense implementation when no don’t-cares are present and an independent audit of implicant-hood and primality at 8, 10 and 12 variables.

We note one consequence of that audit for the wider pipeline. The dense reference minimiser retains only don’t-cares at Hamming distance 1 from a real on-minterm, a documented performance choice; where don’t-cares cluster it therefore under-merges, returning 187 cubes against 522 exact prime implicants on a 10-variable test table, with every extra cube it reports strictly contained in a true prime. Rules obtained from it remain sound, but prime-implicant counts and “essential” designations computed with it are approximations rather than exact Quine–McCluskey quantities.

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